

Selenium	QTP
Open source, free to use and modify, and free of charge	Commercial
Highly extensible	Limited add-ons
Can run tests across different browsers	Can only run tests in Firefox, Internet Explorer and Chrome
Supports various operating systems	Can only be used in Windows
Supports mobile devices	Supports mobile devices using third party software
Can execute tests while the browser is minimised	Needs to have the application under test to be visible on the desktop
Can execute tests in parallel	Can only execute in parallel but using Quality Center, which is a paid for product

Table 3: Advantages of Selenium over QTP

QTP	Selenium
Can test both Web and desktop applications	Can only test Web applications
Comes with a built-in object repository	Has no built-in object repository
Automates faster than Selenium because it is a fully featured IDE	Automates at a slower rate because it does not have a native IDE and only third party IDE can be used for development
Data-driven testing is easier to perform because it has built-in global and local data tables	Data-driven testing is more cumbersome since you have to rely on the programming language's capabilities for setting values for your test data
Can access controls within the browser (such as the Favourites bar, Address bar, Back and Forward buttons, etc)	Cannot access elements outside of the Web application under test
Provides professional customer support	No official user support is being offered
Has native capability to export test data into external formats	Has no native capability to export runtime data onto external formats
Parameterisation support is inbuilt	Parameterisation can be done via programming but is difficult to implement
Test reports are generated automatically	No native support to generate test/bug reports

Table 4: Advantages of QTP over Selenium

that of Selenium RC (Java, Ruby, Python, Perl, PHP or .NET).

The pros of WebDriver are:

- Simpler installation than Selenium RC
 - Communicates directly to the browser
 - Browser interaction is more realistic
 - There's no need for a separate component such as the RC server
 - Faster execution time than Selenium IDE and RC
- The cons are:
- Installation is more complicated than IDE
 - Requires knowledge of programming
 - Needs Selenium RC server to be running
 - API contains redundant and confusing commands
 - Browser interaction is less realistic
 - Inconsistent results and uses JavaScript
 - Slower execution time than WebDriver

Selenium Grid

This is used along with Selenium RC to allow the running of multiple parallel tests across different machines and browsers at the same time. Parallel execution means running different tests at once.

Table 1 indicates how you should go about choosing the most suited tool in Selenium for your needs.

Because of the architectural differences in various Selenium tools, each one supports different versions of a browser and operating systems. This is indicated in Table 2.

Tables 3 and 4 provide a comparison between Selenium and QTP.

So after a comparison, it is evident that QTP has more advanced capabilities in some areas while Selenium is better in the following areas.

- **Cost:** Being open source, everyone is free to use it.
- **Flexibility:** Wide support to various programming languages and browser versions.
- **Parallel testing:** Something that QTP is not able to perform. 

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